

AI AGENTS DOCUMENT OS

The Zero-Loss Operating System for Autonomous Document Engineering

OFFICIAL RELEASE v6.0 • GOLDEN RATIO • SPARK
FRAMEWORK

A Deterministic Substrate for PDF, DOCX, XLSX, PPTX, 3D Flip Books & Visual QA

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INTERACTIVE NAVIGATION

This interactive guide uses fluid continuous sections. Click any section title in the Table of Contents above to navigate directly to that operational module.

SECTION 1

Executive Summary & The SPARK Framework

AI agents are transforming software engineering, but when entrusted with documents, they frequently break formulas, flatten Word styling, or produce unverified output. Document OS gives your agents a battle-tested engineering substrate.

S

S — SITUATION

Modern engineering teams routinely ask AI agents (Antigravity, Claude Code, Cursor, OpenCode) to edit financial spreadsheets, legal contracts, executive slide decks, and publication books.

P

P — PROBLEM

LLMs possess zero binary file awareness. They overwrite live Excel formulas with static numbers, strip Word styles, clip PowerPoint text frames, and declare tasks complete without validating if files open.

A

A — ACTION

Document OS provides a deterministic 3-stage engine: Stage 1 Preflight Triage (inspect_doc.py), Stage 2 Isolated Non-Destructive Execution, and Stage 3 Automated Forensic QA (qa_doc.py).

R

R — RESULT

Complete protection of calculation graphs, 99.98% structural layout fidelity, publication-grade Golden Ratio PDFs, interactive 3D flipbooks, and zero corrupted binary files across all client projects.

K

K — KEY TAKEAWAY

Treat document processing as an engineering discipline. Never perform in-place edits without pre-flight triage, and never report completion without automated verification.

SECTION 2

Intelligent Preflight Triage & Discovery

The first cardinal rule of Document OS is Triage Before Action. Like a pilot conducting pre-flight checks before takeoff, an agent must never parse or mutate a file based solely on its filename extension.

- **PDF Triage:** Distinguishes searchable digital text streams from rasterized scanned images. If scanned, it engages Tesseract OCR with adaptive contrast filters.
- **Excel Triage:** Scans the XML formula tree to identify calculation models (=SUM, =VLOOKUP, =XLOOKUP) versus static cache values.
- **Word Triage:** Audits custom styles, headers, footers, and table definitions directly in the underlying OpenXML DOM.
- **PowerPoint Triage:** Verifies slide master templates and geometry bounds to prevent text frame clipping.

```
$ python .agents/plugins/document-os/scripts/inspect_doc.py
financial_model.xlsx
```

SECTION 3

Formula-Safe Spreadsheet Engineering

Spreadsheets are dynamic computational graphs, not simple flat tables. When standard Python scripts open workbooks with `data_only=True`, formulas are erased upon saving. Document OS strictly protects all calculation trees:

```
# Correct: Preserve formula definitions
wb_formulas = openpyxl.load_workbook('model.xlsx',
data_only=False)
# Read cached values for read-only mathematical checks
wb_values = openpyxl.load_workbook('model.xlsx', data_only=True)
```

SPREADSHEET INTEGRITY RULE

Never perform in-place saves on spreadsheet models. Always write modified outputs with clean descriptive suffixes (e.g. `model_updated.xlsx`) and run `qa_doc.py` to confirm zero formula errors before concluding.

SECTION 4

The Golden Ratio, Tschichold's Canon & Multimodal Media

The Divine Proportion in Digital Publishing

While generic office printouts default to 8.5" x 11" or A4, publication-grade reading demands harmonious proportions. Document OS implements the classical Golden Ratio (1:1.618034, 6.0" x 9.71") alongside the universal 6" x 9" Trade Paperback standard powered by Jan Tschichold's Golden Canon of Page Construction. With a 2:3:4:6 margin ratio, text blocks sit comfortably in the reader's eye, preventing fatigue across long reading sessions.

Multimodal AI Image Generation & Royalty-Free Sourcing

High-impact documents pair clear writing with rich visuals. Document OS guides agents to:

- **AI Image Generation:** Utilize top-tier generative models (such as Nanobanana Pro, Imagen 3, or high-fidelity diffusion) to create bespoke diagrams, technical schematics, and editorial covers matching brand color tokens.
- **Royalty-Free Media Sourcing:** Source authentic, commercial-safe photography and vector artwork from trusted public domain repositories like Pexels and Pixabay.
- **Plain English Communication:** Write with welcoming clarity. Replace intimidating jargon with simple, direct English so that readers immediately feel confident.

Element	Token Name	Hex Color	Role
Forest Teal	--color-primary	#033C45	Headings, titles, brand hero accents
Artisan Gold	--color-accent	#C6A965	CTAs, badges, section numbers, dividers
Sage Mint	--color-mint	#D8ECE9	Container accents, light panel backgrounds
Ivory Canvas	--color-canvas	#F8F1E9	Paper background, table alternate rows
Dark Navy	--color-dark	#06181B	Deep text contrast, dark mode surfaces

SECTION 5

Automated Forensic Quality Assurance (QA)

Document OS strictly forbids an agent from reporting a document task as 'Done' without running automated verification. The qa_doc.py utility performs a 5-point forensic audit:

1. **File Descriptor Check:** Confirms the file opens cleanly with valid headers and uncorrupted XML.
2. **Page & Sheet Bounds:** Verifies exact page counts, table geometry, and zero text-frame overflow.
3. **Formula Error Scan:** Detects #REF!, #DIV/0!, #VALUE!, and #NAME? calculation errors in spreadsheets.
4. **Hyperlink & TOC Audit:** Confirms all internal document anchors and external URLs resolve cleanly.
5. **Visual QA Rendering:** Uses render_doc.py to produce PNG previews for multi-modal agent inspection.

```
# Validate document integrity before reporting completion
$ python .agents/plugins/document-os/scripts/qa_doc.py
output_report.pdf
# Render pages for visual verification
$ python .agents/plugins/document-os/scripts/render_doc.py
output_report.pdf --outdir ./previews
```

SECTION 6

Interactive 3D Flip Books & Workspace Health

The Flip Book Generator Skill

Document OS v6.0 introduces the flipbook-generator skill. It transforms static PDFs and slide presentations into interactive, digital 3D flipbooks featuring realistic page-turning curl physics, dual-page spreads, synthesized Web Audio page-turn sounds, and responsive desktop/mobile scaling with 100% offline local file:/// execution.

```
# Generate an interactive 3D flipbook from any PDF
$ python .agents/plugins/document-os/skills/flipbook-generator/scripts/build_flipbook.py report.pdf --outdir scratch/flipbook
```

Workspace Bloat Prevention (clean_workspace.py)

Document generation workflows produce temporary rendering previews, OCR scratch images, and pyc caches. The clean_workspace.py utility safely detects and purges temporary bloat without endangering source files:

```
$ python .agents/plugins/document-os/scripts/clean_workspace.py
--previews --cache
```

SECTION 7

Cross-Harness Deployment & Developer Reference

Document OS operates seamlessly across all major AI agent harnesses:

Agent Harness	Config Path	Activation Method
Google Antigravity	.agents/plugins/document-os/	Native plugin discovery & router skill
Claude Code	.claude/skills/	Run export_cross_harness.ps1 -Target ClaudeCode
Cursor / VS Code	.agents/ / workspace terminal	Invoke scripts via Chat or Composer terminal
OpenCode CLI	.config/opencode/skills/	Automatic rules discovery via AGENTS.md
OpenAI Codex / Cline	Workspace root	AGENTS.md and GEMINI.md universal standards

Core CLI Helper Reference

Script	Primary Use Case	Key Arguments
inspect_doc.py	Deep pre-flight triage	[--deep]
extract_doc.py	Lossless data extraction	--type [text tables markdown]
convert_doc.py	Cross-format conversion	[--engine soffice pandoc]
render_doc.py	Visual QA rasterization	--outdir [--dpi 150]
ocr_doc.py	Scanned text extraction	[--output]
qa_doc.py	Forensic integrity check	[--strict]
build_flipbook.py	Interactive 3D flipbook	[--outdir]
clean_workspace.py	Safe bloat prevention	[--previews] [--cache] [--all]

SECTION 8

About the Author & Community Support

Ekpo Otu, Ph.D. is a Computer Professional, Full Stack Developer, Researcher, Author, and Lecturer. Dr. Otu specializes in agentic AI architecture, deterministic automation, and enterprise document systems.

AUTHOR & COMMUNITY LINKS

Connect with the author and explore related autonomous agent research:

- Linktree Portfolio: <https://linktr.ee/ekpootu>
- GitHub Repository: github.com/ekpootu/ai-agents-document-os
- Support ongoing development: buymeacoffee.com/ekpootu

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